

TYPES OF ATTACKS

Introduction

- Internet in India is growing rapidly
- Unrestricted no. of free websites, the Internet has deniably opened a new exploitation known as cybercrime
- Activates involve the use of computer, Internet, Cyberspace and WWW
- 1st recorded cybercrime took place in the year 1820
- Indian corporate and government sites have been attacked or defaced more than 780 times between Feb 2002 & Dec 2002
- 3,286 Indian websites were hacked in 5 months – between Jan & June 2009

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- Cyberspace: (by William Gibson in 1984)
 - Worldwide n/w of Computer networks that uses the TCP/IP for communication to facilitate transmission and exchange of data
- Cybersquatting:
 - Means registering, selling or using a domain name with intent of profiting from goodwill of someone else's trademark

Cont...

- Cyberpunk: (by Bruce Bethke, 1980)
 - Mean something like “anarchy via machine” or “machine/computer rebel movement”
- Cyberwarfare:
 - Means information warriors unleashing vicious attacks against an unsuspecting opponent’s computer networks, wreaking havoc and paralyzing nations.
- Cyberterrorism: (by Barry Collin, 1997)
 - Use of disruptive activities, or the threat, against computers and/or networks, with the intention to cause harm or further social, ideological, religious, political or similar objectives or to intimidate any person in furtherance of such objectives

Cybercrime: Definition and Origins of the Word

- “A crime conducted in which a computer was directly or significantly instrumental”
- “Cybercrime is any illegal behavior, directed by means of electronic operations, that targets the security of computer systems and the data processed by them.”
- Computer related crime, Computer crime, Internet crime, E-crime, High-tech crime etc. are synonymous terms

Cont...

- Few definition of Cyber Crime:
 - A crime committed using a computer and the Internet to steal person's identity
 - Crime completed either on or with a computer
 - Any illegal activity done through the Internet or on the computer
 - All criminal activities done using the medium of computers, the Internet, cyberspace and WWW

● Types of attack

- Techno-crime:
 - A premeditated act against a system or systems with the intent to copy, steal, corrupt or otherwise deface or damage part of or the complete computer system
 - Possible when computer connected with the Internet 24 X 7
- Techno-vandalism:
 - These acts of “brainless” defacement of websites and/or other activities, such as copying files and publicizing their contents publicly

Cybercrime and Information Security

- Cybersecurity means protecting information, equipment, devices, computer, computer resource, communication device and information stored therein from unauthorized access, use, disclosure, disruption, modification or destruction

Who are cybercriminals?

- Categorized in 3 groups:

- Type I: Cybercriminals- hungry for recognition
 - Hobby hackers
 - IT professionals
 - Politically motivated hackers
 - Terrorist organizations
- Type II: Cybercriminals- not interested in recognition
 - Psychological perverts
 - Financially motivated hackers
 - State-sponsored hacking
 - Organized criminals

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- Type III: cybercriminals- the insiders
 - Former employees seeking revenge
 - Competing companies using employees to gain economic advantage through damage and/or theft

Classification of Cybercrimes

1. Cybercrime against individual:
 - E-Mail spoofing and other online fraud
 - Phishing
 - Spamming
 - Cyberdefamation
 - Cyberstalking and harassment
 - Computer sabotage
 - Pornographic offenses
2. Cybercrime against property:
 - Credit card frauds
 - Intellectual property crime
 - Internet time theft

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3. Cybercrime against organization:

- Unauthorized accessing of computer
- Password sniffing
- Denial-of-service attacks
- Virus
- E-Mail bombing
- Salami attack
- Logic bomb
- Trojan horse
- Data diddling
- Industrial spying
- Crimes emanating from Usenet newsgroup
- Computer network intrusions
- Software piracy

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4. Cybercrime against society:

- Forgery
- Cyberterrorism
- Web jacking

5. Crimes emanating from Usenet newsgroup:

- Usenet group may carry very offensive, harmful, inaccurate or otherwise inappropriate material or postings that have been misplaced or are deceptive in another way

E-Mail Spoofing

- A spoofed email is one in which e-mail header is forged so that mail appears to originate from one source but actually has been sent from another source

Spamming

- Spamming means sending multiple copies of unsolicited mails or mass e-mails such as chain letters.
- Difficult to control
- In context of “search engine spamming”, spamming is alternation or creation of a document with the intent to deceive an electronic catalog or filing system

Cyberdefamation

- Cognizable offense
- This occurs when defamation takes place with the help of computers and / or the Internet.
- E.g. someone publishes defamatory matter about someone on a website or sends e-mails containing defamatory information.

Internet Time Theft

- The usage of the Internet hours by an unauthorized person which is actually paid by another person
- Comes under hacking

Salami Attack/Salami Technique

- When negligible amounts are removed & accumulated in to something larger. These attacks are used for the commission of financial crimes.

Data Diddling

- This kind of an attack involves altering raw data just before it is processed by a computer and then changing it back after the processing is completed.

Forgery

- Currency notes, revenue stamps, mark sheets etc can be forged using computers and high quality scanners and printers.

Web Jacking

- Hackers gain access and control over the website of another, even they change the content of website for fulfilling political objective or for money.

Industrial Spying/Industrial Espionage

- “Spies” can get information about product finances, research and development and marketing strategies, an activity known as Industrial spying.

Hacking

- Every act committed toward breaking into computer and/or network is hacking
- The purpose of hacking Power, publicity, revenge, adventure, desire to access forbidden information, destructive mindset

Online Frauds

- Spoofing website and E-Mail security alerts, lottery frauds, virus hoax E-Mail

Pornographic offenses

- Child pornography means visual depiction

Software Piracy

- Theft of software through the illegal copying of genuine programs or the counterfeiting and distribution of products intended to pass for the original.
- Illegal copying of programs, distribution of copies of software

Computer Sabotage

- The use of the Internet to hinder the normal functioning of a computer system through the introduction of worms, viruses or logical bombs is referred to as computer sabotage

E-Mail Bombing/ Mail Bombs

- Sending a large no. of E-Mails to the victim to crash victim's E-Mail account or to make victim's server crash

Usenet Newsgroup as the Source of Cybercrimes

- Usenet is a popular means of sharing and distributing information on the web with respect to specific subjects or topic
- Following criminal use Usenet:
 - Distribution/sale of pornographic material
 - Distribution/sale of pirated software package
 - Distribution of hacking software
 - Sale of stolen credit card number
 - Sale of stolen data/stolen property

Computer Network Intrusions

- Crackers can break into computer systems from anywhere in the world and steal data, plant viruses, create backdoors, insert trojan horse or change username and passwords

Password Sniffing

- Programs that monitor and record the name and password of network users as they login at a site

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- Credit Card Frauds
 - Newsgroup Spam/Crimes Emanating from Usenet Newsgroup